# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-SLERP-EN · v2.2 · 2026-11-02 |
| Package | star-slerp |
| Version | 0.1.0 |
| Capability | star.attitude.slerp |
| Maturity | Released & Verified (RELEASED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_slerp |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_slerp-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
| --- | --- |
| star_slerp-0.1.0-py3-none-any.whl | f9c6c73ca8f47c7540b41bd9cd98d6b3362f54620cb034f053edeb02d38c0844 |
| star_slerp.py | 74c8063585c8506e6ebd51060392d9af2b18fc5f83ef5be6d28a689efaabb6be |
| star_slerp-0.1.0-source.zip | d6454bd166159619e20125d3aab0946f7a791bbff17c422d9bb71834f800402c |

# 5. What it does, and its limits

Spherical linear interpolation between unit quaternions along the shorter rotation, and interpolation in a table of attitudes at given times; on 600 pairs of attitudes the largest component difference is 4.4e-16 from SciPy Slerp and 3.3e-16 from a route through NAIF SPICE rotation matrices, axis and angle.

Interpolation at constant rate between two attitudes, nothing else: no angular velocity, no smooth (C1) interpolation across table rows (the rate jumps at each row), no extrapolation, no model of the true motion between samples. For a rotation of exactly 180 degrees both arcs are equally short and the result follows the sign of the q1 supplied; other libraries may choose the other arc. The quaternion convention is scalar-first Hamilton: data in another convention must be converted before use.

- `slerp(q0, q1, t)` returns the attitude reached after the fraction t of the rotation from q0 to q1, about the same axis and at constant rate; quaternions are (w, x, y, z), scalar first, of unit length.
- The path is always the shorter rotation: when q0 and q1 are in opposite hemispheres q1 is negated first. The result is of unit length and in the hemisphere of q0. `interpolate(times, quaternions, t)` applies the same rule between the two rows of the table that bracket t and refuses a t outside the table.
- Measured on 600 pairs of attitudes (rotations from 1e-9 rad to within 0.1 degree of 180 degrees, either hemisphere): the largest component difference is 4.4e-16 from SciPy `Slerp`, 3.3e-16 from a route through NAIF SPICE rotation matrices, axis and angle, and 2.2e-16 from ahrs above its linear-interpolation threshold.
- Every function returns a tuple of four floats or raises `ValueError`: a quaternion that is not four finite numbers of unit length within 1e-9 (never normalised silently), t out of range, a malformed or non-increasing table.

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_slerp_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-slerp}: Interpolation of attitudes along the shortest arc},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_slerp/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts